

Notes: Piskorski, Seru, and Witkin (JF, 2015)

Asset Quality Misrepresentation by Financial Intermediaries: Evidence from the RMBS Market

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1 Big picture

- **Question.** Did financial intermediaries selling nonagency RMBS provide investors false information about collateral quality, where in the securitization chain did that happen, and did it generate economically meaningful investor losses?
- **Setting.** The paper studies privately securitized residential mortgage-backed securities during the housing boom, focusing on loans originated in 2005–2007 and sold through large, reputable underwriters. Investors received detailed loan files and prospectus disclosures, so the paper can compare what was contractually disclosed to what outside data say was true at origination.
- **Clean misrepresentation dimension.** The core object is *second-lien misrepresentation*: a first-lien mortgage is disclosed to investors as if the property has no junior lien, while matched credit-bureau data show a simultaneously originated second mortgage. This means reported combined loan-to-value (CLTV) is materially understated.
- **Core empirical challenge.** The paper must distinguish true false disclosure from ordinary asymmetric information, data-matching error, or later borrower behavior. It also must show that any hidden collateral risk mattered for default and security pricing, and was not just a technical defect with no economic bite.
- **Main contribution.** The paper links three types of evidence:
 - direct loan-level detection of false collateral disclosure using BlackBox loan files matched to Equifax credit-bureau records;
 - institutional evidence from New Century internal files and county deeds records showing lenders knew about the missing second liens;
 - pool- and tranche-level evidence showing that securities backed by more misrepresented collateral suffered worse losses, downgrades, and price declines, yet were not sold at an issuance discount on standard pricing proxies.
- **Main data design.** The raw matched sample contains about **1.9 million** first-lien mortgages originated in 2005–2007. Restricting to loans with nonmissing CLTV and reported CLTV within 1 percentage point of LTV yields **854,959** loans disclosed as having no second lien. The pool-level analysis uses **333** RMBS pools containing **669,462** loans.
- **Headline extent of misrepresentation.** Among loans reported to investors as having no simultaneous second lien, **7.13%** actually have such a lien in the credit-bureau data. If simultaneously originated HELOCs are counted too, the rate rises to **16.64%**. For misrepresented loans, true CLTV exceeds reported CLTV by about **19.57 percentage points** on average.
- **Key descriptive surprise.** Misrepresentation is *not* concentrated in the obvious “junk” corner of the market. About **7.93%** of fully documented loans that report no second lien are misrepresented. Misrepresentation is especially common among **purchase loans (13.75%)** rather than refinances (**3.20%**).
- **Main loan-level result.** A loan with a misreported second lien is about **10.15 percentage points** more likely to become 90 days delinquent within two years of origination, relative to an

otherwise similar loan with truthfully reported no second lien. Given a mean default rate of **14.58%**, that is roughly a **70%** higher default likelihood.

- **Main pricing result.** Lenders charge somewhat higher mortgage rates on misrepresented loans relative to loans with no second lien, but the rate premium is much smaller than on loans with truthfully reported second liens of similar risk. At the RMBS pool level, standard issuance-pricing proxies do *not* show discounts for pools with more hidden second liens.
- **Main “who knew” result.** In the New Century internal file, more than **93%** of loans identified as misreporting second liens are recorded internally as having those liens. In Los Angeles County deed records, more than **93%** of misreported loans have recorded second liens and about **99.67%** have the *same lender* responsible for both first and second liens.
- **Main investor-loss result.** A 1 percentage point increase in the share of misrepresented loans in a pool is associated with about a **0.25 percentage point** increase in cumulative pool losses. The effect is much larger for junior tranches and remains positive even for AAA tranches.
- **Main market-pricing result.** Pools with more misrepresented collateral were *not* issued with meaningfully higher yield spreads or higher AAA subordination levels, even though reported CLTV is strongly priced. This is the paper’s sharpest evidence that investors and rating agencies did not distinguish hidden collateral risk across pools.
- **Economic takeaway.** The paper documents outright misrepresentation in a major debt market, not just standard adverse selection. The hidden risk was large enough to matter for loan defaults, RMBS losses, rating downgrades, and investor price declines, while remaining insufficiently reflected in cross-sectional security pricing.

2 Data and measurement (key objects)

2.1 Why the RMBS market is the right laboratory

- In nonagency RMBS, underwriters and trustees provide investors with detailed loan-level files and summary prospectus disclosure, so the “reported” information set is unusually concrete.
- CLTV and second-lien status are first-order risk variables in mortgage markets because borrower equity is central to default incentives.
- The market involves multiple layers of intermediation—borrower, broker, lender, aggregator, underwriter, rating agency, and institutional investor—which makes it possible for information to be collected inside the chain but not truthfully passed through to end investors.
- Prospectuses also contain representations and warranties, including repurchase obligations if disclosed loan characteristics are materially false. That means the paper is not testing a vague disclosure norm; it is testing a contractually important disclosure margin.

Interpretation. This setting allows the authors to move beyond “investors knew less than issuers” and ask the harder question: were investors given *false statements* about observable collateral characteristics?

2.2 Institutional structure of the supply chain

Figure 1 in the paper gives the core map. The flow is:

1. lenders originate mortgages, often through brokers;
2. lenders or financial intermediaries pool loans into securitization trusts;
3. underwriters structure and sell RMBS tranches;
4. rating agencies and institutional investors rely on disclosed collateral data;
5. contractual representations and warranties are meant to guarantee that those disclosures are accurate.

The paper emphasizes two information sets:

- the information collected within the financial industry during origination and securitization;
- the narrower information ultimately disclosed to RMBS investors in loan tapes and prospectuses.

Interpretation. The entire paper is about the wedge between these two information sets.

2.3 BlackBox loan-level data

The first core data source is **BlackBox Logic**, a private data set with roughly **21 million** privately securitized mortgages originated after 1999. It covers about **90%** of privately securitized subprime, Alt-A, and prime mortgages from that period.

The BlackBox data contain both:

- **static origination variables:** origination date, balance, FICO, interest rate, term, mortgage type, CLTV, occupancy status, and other underwriting characteristics;
- **dynamic performance variables:** balances, payments, and delinquency status over time.

Most importantly, BlackBox reflects the loan-level information that servicers and trustees provide to securitization investors. That makes it the paper's measure of what investors were told.

2.4 Equifax credit-bureau data and the merge

The second core data source is **Equifax**, which records borrower-level credit histories and debt obligations from consumer credit files. Banks rely heavily on such data to screen borrowers, so these records are treated as highly accurate.

Equifax matched its records to BlackBox using a proprietary algorithm based on more than 25 variables. The paper imposes a very conservative restriction:

- keep only loans with the **highest Equifax merge-confidence level**;
- focus on loans originated in **2005–2007**, the years for which this high-confidence merge is available;

- restrict attention to loans disclosed as **first liens**.

After these restrictions, the base matched sample contains about **1.9 million loans**.

Interpretation. The paper willingly gives up sample size to make the “this was the same loan in both databases” objection as weak as possible.

2.5 Constructing the second-lien misrepresentation measure

The main measure is built in two stages.

First, the authors identify loans that are reported to investors as having *no* second lien. They keep loans with:

$$\text{Reported No Second}_i = 1 \left\{ \text{CLTV}_i^{\text{reported}} \text{ is nonmissing and } \left| \text{CLTV}_i^{\text{reported}} - \text{LTV}_i^{\text{reported}} \right| \leq 1 \right\}. \quad (1)$$

This leaves **854,959** loan files.

Second, among those loans, they classify a loan as misrepresented if Equifax shows a simultaneously originated second mortgage:

$$\text{Misreported Second}_i = 1 \left\{ \text{Reported No Second}_i = 1 \text{ and Equifax shows a second lien within 45 days of origination} \right\}. \quad (2)$$

Key implementation choices:

- the 45-day window allows for minor recording-date differences;
- the main definition excludes HELOCs, even though many prospectuses include them in CLTV, so the benchmark estimate is conservative;
- loans with missing CLTV are *not* counted as misrepresented, again making the estimate a lower bound.

Interpretation. The measure is deliberately narrow. The authors are not trying to capture every possible bad disclosure practice; they are isolating one hard, verifiable, economically important lie.

2.6 Auxiliary data sets used to validate and extend the analysis

The paper supplements BlackBox-Equifax with several additional sources:

- **New Century internal loan files** obtained through IP Recovery after the lender’s bankruptcy. These reveal what the lender itself recorded at origination.
- **Los Angeles County deed records** from First American. These show legally recorded liens and lender identities for the property.
- **Bloomberg ABS data** on pool and tranche losses, ratings downgrades, subordination levels, and price changes as of 2014Q1.

- **Insurance-company holdings** from Merrill et al. (2014), which identify one important class of institutional investors holding specific RMBS tranches.

2.7 Main outcome variables

The paper studies four classes of outcomes.

Loan-level performance.

$$\text{Default}_i = 1 \tag{3}$$

if loan i ever goes **90 days past due within two years of origination**.

Loan-level pricing.

$$r_i = \text{mortgage interest rate at origination.} \tag{4}$$

Pool-level outcomes.

- cumulative pool loss up to 2014Q1;
- issuance yield spread;
- AAA subordination level.

Tranche-level outcomes.

- cumulative losses by rating bucket;
- subsequent ratings downgrades;
- subsequent price changes.

2.8 Descriptive facts that matter

The descriptive results already foreshadow the paper’s argument:

- **7.13%** of loans that report no second lien are misrepresented;
- with HELOCs counted, the figure jumps to **16.64%**;
- mean reported CLTV for the full sample is about **78.93%**, while the true CLTV of misrepresented loans is roughly **19.57 percentage points** higher than disclosed;
- misrepresented loans are disproportionately **purchase mortgages** (about **71%** of misrepresented loans versus **36%** in the overall no-second sample);
- misrepresentation remains substantial in **fully documented loans**.

Interpretation. The paper’s central point appears immediately: hidden second liens are not a niche low-doc anomaly. They are a meaningful and broadly distributed distortion of collateral information.

3 Models and empirical specifications (all equations + interpretation)

3.1 Cross-sectional patterns in misrepresentation

The first empirical exercise asks which observable loan characteristics predict second-lien misrepresentation:

$$\text{Misreported Second}_i = \alpha + X_i' \beta + \varepsilon_i, \quad (5)$$

where X_i includes FICO, origination balance, reported CLTV, purpose dummies, documentation status, ARM indicators, origination-time fixed effects, and in some specifications state and underwriter fixed effects.

Interpretation.

- This is not the main causal test. It is a diagnostic map of where misrepresentation shows up.
- If misrepresented loans look like loans that genuinely use second liens, that supports the idea that the lender probably knew these were piggyback structures.
- Underwriter fixed effects let the paper separate loan-composition differences from persistent underwriting heterogeneity.

3.2 Loan default regression

The paper then asks whether hidden second liens matter for ex post loan risk:

$$\text{Default}_i = \alpha + \gamma \text{Misreported Second}_i + X_i' \beta + \varepsilon_i. \quad (6)$$

The default indicator equals one if the loan goes 90 days past due at least once in the first two years after origination.

The control vector contains variables disclosed to investors and commonly used in the mortgage default literature, including reported FICO, LTV, mortgage characteristics, and origination cohort controls. In expanded specifications, the paper also uses squared and cubed terms for FICO and LTV and adds state fixed effects.

Interpretation. The coefficient γ measures whether two loans that *look the same to investors* perform differently because one had an undisclosed second lien.

3.3 Expanded loan-status comparison

To compare misreported loans not just with “no-second” loans but also with truthfully disclosed second-lien loans, the paper estimates:

$$\text{Default}_i = \alpha + \beta_1 M_i^{<100} + \beta_2 R_i^{<100} + \beta_3 M_i^{\geq 100} + \beta_4 R_i^{\geq 100} + X_i' \Gamma + \varepsilon_i, \quad (7)$$

where:

- $M_i^{<100}$ is a misreported second lien with true CLTV < 100 ,
- $R_i^{<100}$ is a truthfully reported second lien with CLTV < 100 ,
- $M_i^{\geq 100}$ is a misreported second lien with true CLTV ≥ 100 ,
- $R_i^{\geq 100}$ is a truthfully reported second lien with CLTV ≥ 100 .

The omitted category is loans that truthfully report no second lien.

Interpretation. This is the paper’s cleanest “horse race.” If misreported loans default like truthfully reported second-lien loans, then the hidden second lien is capturing economically real risk rather than random measurement noise.

3.4 Mortgage-rate regression

The same logic is applied to loan pricing:

$$r_i = \alpha + \delta \text{Misreported Second}_i + X_i' \beta + \varepsilon_i, \quad (8)$$

or in expanded form using the same four second-lien/CLTV buckets as above.

Interpretation.

- If lenders were totally unaware of the second lien, mortgage rates on misrepresented loans should look like rates on truthful no-second-lien loans.
- If lenders were partly aware, rates should be somewhat higher.
- If disclosed pricing to investors fully reflected true risk, misreported loans should price like truthfully reported second-lien loans of comparable CLTV.

The paper finds the middle case: some private awareness by lenders, but not full pass-through into the terms that investors see.

3.5 Pool-level RMBS regressions

To move from loans to securities, the paper estimates:

$$Y_p = \alpha + \gamma \text{Percent Misreported Second}_p + X_p' \beta + \varepsilon_p, \quad (9)$$

where p indexes RMBS pools.

Three different dependent variables are studied:

1. cumulative pool losses,
2. issuance yield spread,
3. AAA subordination level.

The pool-level controls include means of disclosed origination characteristics such as reported CLTV, mean FICO, fraction low documentation, fraction purchase loans, and overcollateralization. Some specifications also include underwriter fixed effects.

Interpretation.

- When Y_p is *losses*, the regression asks whether hidden second liens predict ex post RMBS damage.
- When Y_p is *yield spread* or *subordination*, the regression asks whether investors and rating agencies priced this hidden risk at issuance.

3.6 Tranche-level loss, downgrade, and price regressions

The paper then runs versions of the pool specification using tranche outcomes as the dependent variable:

$$Y_{pt} = \alpha + \gamma \text{Percent Misreported Second}_p + X_p' \beta + \varepsilon_{pt}, \quad (10)$$

where t indexes tranches within pool p .

Outcomes include:

- cumulative losses on below-AAA tranches,
- cumulative losses on below-A tranches,
- cumulative losses on AAA tranches,
- downgrades of AAA tranches in notches,
- price changes of AAA tranches.

Interpretation. Because RMBS tranches are highly levered claims on pool cash flows, even a modest increase in pool losses can destroy junior-tranche value and still spill into senior tranches.

3.7 Insurance-company holdings tests

Using holdings data for insurance companies, the paper estimates whether pools with more hidden second liens were:

- less likely to be purchased by insurers,
- associated with larger subsequent losses on insurer-held pools or tranches,
- associated with larger price declines on insurer-held AAA tranches.

Interpretation. This is not a complete investor census, but it is a direct test for one major institutional buyer class.

3.8 Estimation choices

A few implementation choices matter:

- the paper uses **OLS** even when the dependent variable is binary;
- many specifications include time, state, or underwriter fixed effects;
- some standard errors are clustered at the state level;
- the Internet Appendix reports similar inferences using nonlinear models such as probit.

Interpretation. The paper is mostly about transparent partial-correlation evidence with rich controls and validation, not about a single quasi-experimental shock.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why this misrepresentation measure is unusually credible

The strongest part of the paper is that the variable of interest is not inferred from poor performance alone. It is constructed from a direct comparison between:

- what investors were told in the securitization file, and
- what the banking system’s own credit-bureau data show about simultaneous liens.

This is important for three reasons:

1. CLTV is a core risk characteristic, not a peripheral field.
2. Simultaneous second liens are objectively verifiable.
3. Prospectuses often explicitly say CLTV includes subordinate liens, so omission is naturally interpreted as false disclosure rather than mere ambiguity.

4.2 How the paper addresses match-quality concerns

The obvious threat is that BlackBox and Equifax records were incorrectly merged. The paper pushes hard against that concern.

Evidence for high-quality matching:

- it keeps only the highest-confidence Equifax matches;
- origination balance, origination date, and payment histories line up closely across the two data sets;
- only about **0.3%** of the highest-confidence matches show different payment status.

Interpretation. The paper turns a generic “bad match” critique into a concrete empirical question and shows the answer is “probably not.”

4.3 Why the default results are not just a mechanical artifact

The paper adds three separate validation arguments.

First, economic sign test. If the match were random, identified “misrepresentations” would not line up so strongly with higher delinquency. Instead, the paper finds exactly the default pattern mortgage theory predicts for hidden leverage.

Second, placebo on likely bad matches. The Internet Appendix studies the small subset of observations where first-mortgage balances do not match exactly across data sets. Those records are the ones most likely to be mismatched. They do *not* show the same adverse performance pattern as the misreported-second-lien loans.

Third, external validation. Loans flagged as misrepresented by the BlackBox-Equifax merge are cross-checked in New Century internal data and county deeds records, where more than **93%** are confirmed.

4.4 Why the paper believes misrepresentation occurred within the financial industry

The evidence in Section V is the closest thing in the paper to a “who did it” design.

- In the New Century internal file, more than **93%** of loans identified as misreporting second liens are recorded internally as having those liens.
- In Los Angeles County deeds records, more than **93%** of such loans show legally recorded second liens.
- For about **99.67%** of those deed-validated cases, the same lender originated both first and second liens.
- For more than **20%** of misreported second-lien loans in that county sample, the lender and underwriter are the same institution.

Interpretation. This makes a pure “the borrower hid it from the financial industry” story very hard to sustain for second-lien misrepresentation.

4.5 Why the pricing results are informative about investor sophistication

The pool-pricing evidence is subtle. The paper does *not* observe every actual issuance transaction price. Instead, it uses two standard pricing proxies:

- imputed issuance yield spread,
- AAA subordination level.

The credibility logic is comparative:

- reported CLTV is strongly related to both pricing measures, so the proxies are responsive to disclosed collateral risk;
- hidden second-lien misrepresentation is strongly related to subsequent losses;
- yet the misrepresentation share is not significantly related to those issuance-pricing proxies.

Interpretation. Investors and rating agencies seem to have priced what they were told, but not the hidden risk that was not truthfully disclosed.

4.6 What the paper can and cannot cleanly identify

What it identifies well:

- the existence of substantial false disclosure on a hard collateral margin;
- the fact that such loans are riskier ex post;
- that this hidden risk translated into larger RMBS losses;
- that cross-pool hidden risk was not adequately reflected in standard issuance-pricing measures.

What it does not fully pin down:

- the exact legal or organizational point where every misrepresentation was introduced;
- actual negotiated issuance prices for every tranche;
- holdings for every investor class;
- the full universe of possible misrepresentations, since the main measure is intentionally narrow.

4.7 Why the paper's estimates are conservative lower bounds

The authors argue their main estimates are lower bounds for at least four reasons:

- they examine only one easy-to-quantify collateral margin in the main analysis;
- they exclude loans with missing CLTV from the misrepresentation count;
- they exclude HELOCs from the main second-lien definition;
- some second liens may not have been reported to Equifax promptly, so even the credit-bureau comparison can miss some hidden seconds.

This lower-bound logic is reinforced later by a second misrepresentation measure based on occupancy status.

4.8 Broader equilibrium and welfare interpretation

The paper is careful not to claim a full welfare calculation, but it pushes the discussion beyond pure accounting fraud.

- Second-lien misrepresentation alone could affect mortgages with balances of up to about **\$100 billion**.
- A simple back-of-the-envelope implies potential intermediary losses from repurchases on the order of **tens of billions of dollars**, with one illustration around **\$45 billion**.
- Once investors learn that hidden collateral quality is widespread, a lemons-style market shut-down becomes possible. The authors explicitly connect this to the collapse of the nonagency RMBS market.

Interpretation. The paper’s contribution is not just “there was fraud” but “fraud on this margin can meaningfully distort capital allocation in a large debt market.”

5 Tables and figures

5.1 Figure 1: the securitization chain and information flow

Read:

- Figure 1 is a schematic, not a statistical result.
- It lays out the chain from borrower and lender to underwriter, rating agency, and final investor.
- The key visual message is that information is collected early in the chain, then repackaged and certified before sale.

Economic message. The figure makes the paper’s central mechanism intuitive: a variable like second-lien status can be known inside the chain but omitted from the investor-facing disclosure.

5.2 Table I: how much misrepresentation is there?

Read:

- Overall, **7.13%** of loans reported as having no simultaneous second lien are misrepresented.
- Including HELOCs raises the figure to **16.64%**.
- Misrepresented loans understate true CLTV by about **19.57 percentage points**.
- Misrepresentation among fully documented loans is still **7.93%**.
- The purchase-loan rate is **13.75%**, versus only **3.20%** for refinance loans.

Economic message. Hidden second liens are common, large in risk terms, and not confined to the obvious low-documentation part of the market.

5.3 Table II: which loans are more likely to be misrepresented?

Read:

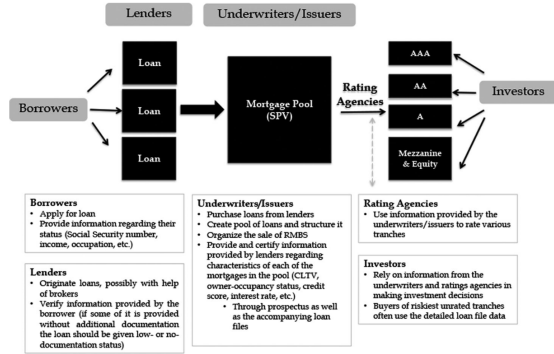


Figure 1. Key players involved in the creation and sale of RMBS. This figure presents a simple schematic of key players involved in the creation and sale of assets in this market. We also discuss the associated information flow.

Table I
Descriptive Statistics and Average Second-Lien Misrepresentation Level

Panel A presents summary statistics of key variables for mortgages reported as having no simultaneous second liens to the RMBS trustee (loans reported as such in the BlackBox data set). The sample comprises these loans merged with a high level of confidence with the credit bureau data. Panel A also shows summary statistics for the subsample of these mortgages that consists of loans with misreported second liens. *FICO* is the borrower's FICO credit score at loan origination. *Balance* is the initial loan balance (in thousands of dollars). *CLTV* is the loan's origination combined loan-to-value ratio in percentage terms reported to investors. *No Cash Out Refi* and *Cash Out Refi* are dummies that take a value of one if the loan purpose was a no-cash-out refinancing or cash-out refinancing, respectively, and zero otherwise. *Low or No Doc.* is a dummy that takes a value of one if the loan was originated with no or limited documentation, and zero otherwise. *ARM* and *Option ARM* are dummies that take a value of one if the loan type was an ARM or option ARM, respectively, and zero otherwise. Panel B presents the percent of loans reported as having no simultaneous second liens that are misrepresented (loans identified as having simultaneous second liens based on our method). Column (1) shows results for the overall sample, column (2) for the subsample of fully documented loans, column (3) for the subsample of loans used to purchase homes, and column (4) for the subsample of loans used to refinance existing mortgages, while column (5) shows results when we expand the definition of simultaneous second liens to HELOCs originated simultaneously with the first-lien mortgage.

Panel A: Descriptive Statistics for Mortgages Reported as Having No Simultaneous Second Liens

	Overall sample		Loans with misreported second liens	
	Mean	SD	Mean	SD
FICO	663.5	76.90	665.7	62.22
Balance	293.1	238.9	242.19	187.8
CLTV	78.93	9.95	79.98	4.96
Purchase	0.36	0.48	0.71	0.45
No Cash Out Refi	0.12	0.32	0.08	0.28
Cash Out Refi	0.50	0.50	0.19	0.39
Low or no doc.	0.51	0.50	0.45	0.49
ARM	0.47	0.49	0.70	0.45
Option ARM	0.16	0.37	0.07	0.25
Number of loans	854,959		60,965	

Panel B: Level of Misrepresented Second Liens

	Overall sample (1)	Fully documented (2)	Purchase loans (3)	Refinance loans (4)	Overall sample (with HELOCs) (5)
Percent misreported	7.13%	7.93%	13.75%	3.20%	16.64%

Difference between actual CLTV and reported CLTV for loans with misreported seconds: Mean = 19.57% (*SD* = 8.00%).

Table II
Second-Lien Misrepresentation across Borrower and Loan Characteristics

This table presents OLS estimates from regressions in which the dependent variable takes a value of one if the loan is *Misreported Second Lien* and zero otherwise. The controls represent the values reported to investors (the corresponding values in the BlackBox database). Column (2) adds fixed effects corresponding to the loan origination time, with 2005 being the omitted category. Column (3) adds fixed effects corresponding to the location of the property backing the loan. Column (4) adds fixed effects capturing the identity of the underwriter that sold a loan. Column (5) reports standard errors clustered at the state level. All estimates are in percentage terms; standard errors are in parentheses. **p* < 0.10, ***p* < 0.05, and ****p* < 0.01. Variables are defined in Table I.

	(1)	(2)	(3)	(4)	(5)
FICO	0.00449*** (0.000467)	0.00523*** (0.000471)	0.00454*** (0.000472)	0.00670*** (0.000481)	0.00670*** (0.00135)
Balance	-0.00810*** (0.000123)	-0.00773*** (0.000124)	-0.00751*** (0.000131)	-0.00769*** (0.000131)	-0.00769*** (0.000783)
CLTV	-0.163*** (0.00305)	-0.161*** (0.00305)	-0.165*** (0.00308)	-0.161*** (0.00307)	-0.161*** (0.0643)
No Cash Out Refi	-8.202*** (0.0917)	-7.990*** (0.0920)	-8.176*** (0.0924)	-7.827*** (0.0919)	-7.827*** (0.713)
Cash Out Refi	-12.03*** (0.0646)	-11.93*** (0.0646)	-11.71*** (0.0653)	-11.37*** (0.0651)	-11.37*** (0.755)
Low or no doc.	-1.501*** (0.0593)	-1.504*** (0.0593)	-1.373*** (0.0596)	-1.023*** (0.0596)	-1.023*** (0.285)
ARM	6.408*** (0.0641)	6.397*** (0.0640)	6.439*** (0.0645)	5.140*** (0.0670)	5.140*** (0.337)
Option ARM	-2.797*** (0.120)	-2.630*** (0.125)	-2.358*** (0.125)	-1.373*** (0.128)	-1.373*** (1.114)
Originated in 2006H1		2.213*** (0.0745)	2.256*** (0.0744)	2.030*** (0.0764)	2.030*** (0.233)
Originated in 2006H2		-0.444*** (0.0796)	-0.422*** (0.0796)	-0.538*** (0.0810)	-0.538*** (0.321)
Originated in 2007H1		-1.820*** (0.0978)	-1.861*** (0.0978)	-2.252*** (0.101)	-2.252*** (0.418)
Originated in 2007H2		-3.993*** (0.262)	-4.028*** (0.262)	-4.048*** (0.262)	-4.048*** (0.406)
State fixed effects	No	No	Yes	Yes	Yes
Underwriter fixed effects	No	No	No	Yes	Yes
SEs clustered by state	No	No	No	No	Yes
Number of loans	854,959	854,959	854,959	854,959	854,959
Percent misrepresented	7.131	7.131	7.131	7.131	7.131
R ²	0.0646	0.0668	0.0712	0.0861	0.0861

- Misrepresented loans tend to have **higher FICO**, **smaller balances**, and **lower reported CLTV**.
- They are more likely to be **purchase loans** and less likely to be refinances.
- They are **less** likely to be low- or no-documentation loans.
- They are more likely to be ARMs and less likely to be option ARMs.
- The time fixed effects show misrepresentation rising into the boom and peaking around **2006H1**, then falling as the market tightened.

Economic message. Misrepresented loans look like classic piggyback-loan structures, which supports the idea that lenders understood the true financing stack even when investors did not.

5.4 Table III and Figure 2: do misrepresented loans default more, and were they priced as riskier?

Table III
Second-Lien Misrepresentation, Loan Default, and Mortgage Interest Rate

Panel A presents OLS estimates from regressions in which the dependent variable takes a value of one if the mortgage ever defaults (ever goes 90 days past due on payments) in the first two years since origination, and zero otherwise. Panel B presents OLS estimates from regressions in which the dependent variable is the mortgage's interest rate at origination. Column (1) shows the results with *Misreported Second Lien* as a control variable. The sample used in these regressions consists of loans that reported to investors no presence of second liens. Columns (2) to (4) show the results when we extend the sample to include loans that reported second liens to investors. These specifications use *Misreported Second* $\times$ *CLTV* ≥ 100 , *Misreported Second* $\times$ *CLTV* < 100 , *Reported Second* $\times$ *CLTV* ≥ 100 , and *Reported Second* $\times$ *CLTV* < 100 as control variables, where the excluded category comprises loans that truthfully reported no second liens. The *CLTV* term (computed from credit bureau data) in these interactions takes a value of one if the loan has a CLTV ratio in the appropriate range, and zero otherwise. "Other Controls" include origination variables reported to investors used in Table III such as FICO, interest rates, LTV ratios, and the origination time fixed effects. Squared and cubed terms for FICO and LTV ratios are also included to account for potential nonlinear effects. The estimates are in percentage terms; standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Panel A: Mortgage Default and Misreported versus Reported Second Lien				
<i>Misreported second</i>	10.15*** (0.1442)			
<i>Misreported Second</i> $\times$ <i>CLTV</i> < 100		5.725*** (0.2791)	6.404*** (0.2776)	6.404*** (0.4841)
<i>Reported Second</i> $\times$ <i>CLTV</i> < 100		6.252*** (0.1197)	6.636*** (0.1192)	6.636*** (0.9360)
<i>Misreported Second</i> $\times$ <i>CLTV</i> ≥ 100		10.19*** (0.1943)	10.97*** (0.1934)	10.97*** (1.3816)
<i>Reported Second</i> $\times$ <i>CLTV</i> ≥ 100		13.26*** (0.1091)	14.02*** (0.1091)	14.02*** (2.3059)
Other controls	Yes	Yes	Yes	Yes
State fixed effects	No	No	Yes	Yes
SEs clustered by state	No	No	No	Yes
Number of loans	854,959	1,109,250	1,109,250	1,109,250
Percent default	14.58	17.10	17.10	17.10
Percent misrepresented	7.131	5.212	5.212	5.212
R^2	0.160	0.162	0.172	0.172
Panel B: Interest Rates at Origination and Misreported Versus Reported Second Liens				
<i>Misreported second</i>	0.0875** (0.00536)			
<i>Misreported Second</i> $\times$ <i>CLTV</i> < 100		-0.00237 (0.00988)	-0.00358 (0.00994)	-0.00358 (0.0325)
<i>Reported Second</i> $\times$ <i>CLTV</i> < 100		0.121*** (0.00427)	0.124*** (0.00426)	0.124*** (0.0392)
<i>Misreported Second</i> $\times$ <i>CLTV</i> ≥ 100		0.0956*** (0.00693)	0.0882*** (0.00689)	0.0882 (0.0848)

(Continued)

Table III—Continued

Panel B: Interest Rates at Origination and Misreported Versus Reported Second Liens				
	(1)	(2)	(3)	(4)
<i>Reported Second</i> $\times$ <i>CLTV</i> ≥ 100		0.295*** (0.00389)	0.290*** (0.00389)	0.290** (0.125)
Other controls	Yes	Yes	Yes	Yes
State fixed effects	No	No	Yes	Yes
SEs clustered by state	No	No	No	Yes
Number of loans	854,959	1,109,250	1,109,250	1,109,250
Mean interest rate	6.458	6.654	6.654	6.654
Percent misrepresented	0.0713	5.212	5.212	5.212
R^2	0.754	0.677	0.682	0.682

Read:

- In Table III, Panel A, the baseline misreported-second coefficient is about **10.15 percentage points** in default probability.
- Relative to truthful no-second-lien loans, both misreported and truthfully reported second-lien loans default much more, especially when actual CLTV is ≥ 100 .

- Figure 2 shows that over the first two years of loan life, cumulative default rates for misrepresented-second loans track closely those of truthfully reported second-lien loans, and both lie well above truthful no-second-lien loans.
- In Table III, Panel B, misrepresented-second loans carry only a small mortgage-rate premium over truthful no-second-lien loans—about **9 basis points** in the simplest comparison.
- By contrast, truthfully reported second-lien loans receive materially larger markups, roughly **12 bps** when $CLTV < 100$ and about **29 bps** when $CLTV \geq 100$.

Economic message. Hidden second liens are economically real risk, but that risk is only partially reflected in loan pricing and not priced like openly disclosed second liens.

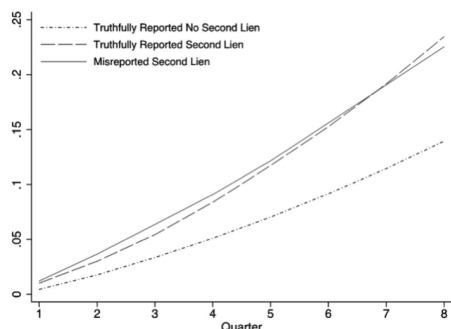


Figure 2. Misrepresentation and loan cumulative default rates. This figure shows the estimated cumulative default rates in the first two years since origination at the loan level based on second-lien status, holding constant at their overall sample mean all other observables used in Table II such as FICO. The y-axis is the cumulative percentage of loans that were 90 days past due at least once.

Table IV
Where Did Misrepresentation Occur? Evidence from New Century Loans and the Registry of Deeds

This table examines a sample of internal banking records for New Century and the Los Angeles County deeds records merged with records from the Black Box-Equifax database. The first row of Panel A examines the sample of New Century loans identified as having a misrepresented second lien and shows the percentage of these loans for which the second lien is correctly reported in the New Century internal data. The second and third rows of Panel A examine a sample of loans originated in L.A. County that misrepresented second liens to investors and show, respectively, the percentage of these loans that have a second lien reported in the deeds records and the percentage of these loans that have the same lender responsible for the first and second liens. Panel B shows the top three underwriters associated with the main lenders responsible for the misrepresented second lien loans in L.A. County. These lenders account for around 80% of lending in Los Angeles County. The numbers in parentheses show what percentage of loans of a given lender was sold by a given underwriter.

Panel A: Misrepresentations in the New Century Internal Database and L.A. County Registry of Deeds

	Percentage	Count
Loans with misrepresented second lien being reported as having a second lien in New Century data set	93.70%	1,279
Loans with misrepresented second lien being reported as having a second lien in the L.A. county deeds records	93.70%	2,978
Loan with misrepresented second lien has the same lender responsible for first and second liens	99.67%	2,978

Panel B: Lenders and Underwriters Associated with Misreported Second Lien Loans in Los Angeles County

Lenders	Top 3 Underwriters
Countrywide	Countrywide (40.3%); Bear Stearns (8.9%); UBS (8.5%)
WMC	Merrill Lynch (51.8%); Bank of America (29.8%); Morgan Stanley (16%)
Lehman Brothers	Lehman Brothers (98.6%); Merrill Lynch (0.6%); Barclays (0.3%)
New Century	Lehman Brothers (42.9%); Citigroup (25%); Goldman Sachs (15.2%)
First Franklin	Lehman Brothers (94.6%); Bear Stearns (1.34%); HSBC (1.34%)
MortgageIT	Lehman Brothers (84%); Bear Stearns (1.9%); Morgan Stanley (1.9%)
DB Home Lending	Deutsche Bank (100%)
Peoples Choice	Lehman Brothers (100%)
Wells Fargo	Bear Stearns (31%); Lehman Brothers (22.6%); Citigroup (20.2%)
NBGI	Deutsche Bank (32.5%); UBS (26.5%); Lehman Brothers (10.8%)
Greenpoint	Merrill Lynch (42.1%); Morgan Stanley (36.8%); Greenpoint (4%)
Resmae	Deutsche Bank (93.3%); J.P. Morgan (2.7%); Lehman Brothers (2.7%)
Mandalay	Lehman Brothers (74.6%); Citigroup (23.7%); Goldman Sachs (1.7%)
JP Morgan Chase	J.P. Morgan (93.8%); Lehman Brothers (4.3%); Countrywide (2.1%)
Metrocities	Morgan Stanley (45%); Deutsche Bank (12.5%); Lehman Brothers (10%)
Bank of America	Bank of America (89.5%); Bear Stearns (7.9%); RBS (2.6%)

5.5 Table IV, Figure 3, and Figure 4: where in the chain did misrepresentation occur?

Read:

- Table IV, Panel A shows more than **93%** confirmation of misrepresented second liens in both New Century internal files and Los Angeles deed records.
- In the deed data, about **99.67%** of validated cases have the same lender on the first and second liens.
- Figure 3 shows that lenders such as Countrywide and New Century account for outsized shares of misrepresented second-lien loans relative to their market shares.

- Table IV, Panel B links major lenders to multiple major underwriters, including Lehman Brothers, Merrill Lynch, Deutsche Bank, Bear Stearns, Citigroup, and others.
- Figure 4 shows substantial underwriter heterogeneity. The omitted category, Credit Suisse, has about **3.37%** misrepresentation, while Lehman Brothers is roughly **five times** higher after controlling for loan characteristics.
- Yet every underwriter in the figure has a misrepresentation level significantly above zero.

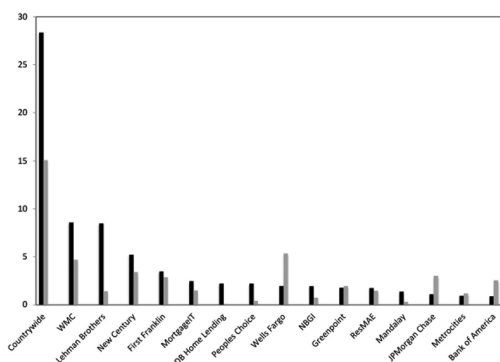


Figure 3. Second-lien misrepresentation across lenders in the Los Angeles county sample. The black bars present the distribution of misreported second-lien loans (in percent) among main lenders in the sample of misreported second lien loans originated in Los Angeles County. The sample is based on Black Box-Equifax data provided by the credit bureau, merged with Los Angeles County deeds records. The gray bars show the overall distribution of second-lien loans among 16 lenders in the Los Angeles County deeds records during 2005 and 2006.

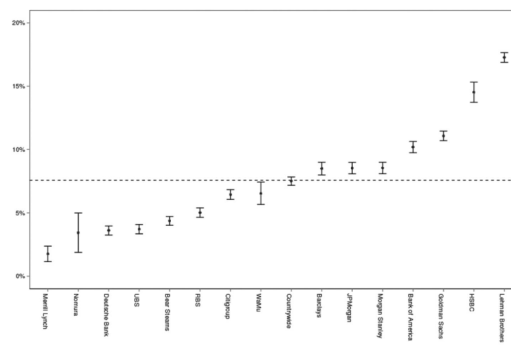


Figure 4. Misrepresentations across underwriters. This figure plots the percentage of loans with misreported second liens by underwriter along with a 95% confidence interval. Coefficients result from adding underwriter fixed effects to the specifications in Table II. These levels are obtained by adding each underwriter fixed effect to the level of misrepresentation for the omitted category (Credit Suisse) with other covariates at their means. The dashed straight line shows the mean misrepresentation level.

Economic message. The lender almost surely knew in many cases, the underwriter often could have known, and misrepresentation was a market-wide phenomenon rather than a one-firm scandal.

5.6 Table V: pool losses, tranche losses, issuance pricing, downgrades, and price declines

Read:

- In Panel A, a 1 percentage point increase in the pool's misrepresentation share is associated with about a **0.25 percentage point** increase in cumulative pool losses.
- In the same panel, the misrepresentation share is statistically insignificant for both **issuance yield spread** and **AAA subordination level**.
- Reported CLTV, by contrast, is strongly priced: a 10 percentage point increase in reported CLTV raises yield spread by about **33 bps** and AAA subordination by about **1.7 percentage points**.
- Panel B shows much larger effects on junior tranches: roughly **1.4 percentage points** more below-AAA loss and **5.7 percentage points** more below-A loss for each 1 percentage point increase in hidden-second share.
- Even AAA tranches are hit: about **0.21 percentage points** more cumulative loss, larger downgrades, and larger subsequent price declines.

- At the sample mean misrepresentation rate of **4.3%**, the implied extra losses are about **0.9%** for AAA tranches, **6%** for below-AAA, and **24%** for below-A tranches.

Economic message. Hidden collateral risk mattered a lot ex post, but cross-pool variation in that hidden risk was not disciplined by standard issuance pricing or rating protection.

Table V
Impact on RMBS Market: Second-Lien Misrepresentation, Losses, and Pricing

Panel A presents OLS estimates from regressions in which the dependent variable is (i) the pool's cumulative loss, defined as the initial balance weighted average cumulative loss of pool tranches (columns (1) and (2)); (ii) the pool's issuance yield spread in percentage terms (columns (3) and (4)); and (iii) the pool's AAA subordination level in percentage terms, defined as the lowest subordination level given to a tranche in the pool that received a credit rating of AAA (columns (5) and (6)). Panel B presents OLS estimates from regressions from the sample of 333 pools in which the dependent variable is (i) the pool cumulative loss on tranches with below-AAA initial rating by Standard & Poor's; (ii) the pool cumulative loss on tranches with below-AAA initial rating by Standard & Poor's; (iii) the pool cumulative loss on tranches with below-A initial rating by Standard & Poor's; (iv) the pool cumulative loss on tranches with AAA initial rating by Standard & Poor's; (v) the ratings downgrades of AAA tranches in notches in the sample of 2,553 tranches (the more negative the number, the larger the downgrade), and (vi) the change in the price of AAA tranches in percentage terms in the sample of 1,706 tranches for which Bloomberg reports these data. *Percent Misreported* is the percentage of loans in the pool identified as having a second-lien misrepresentation. *Reported CLTV* is the average reported CLTV of loans in the pool disclosed to investors. Other controls include the pool-level means of origination variables such as the credit score (FICO) of borrowers or the fraction of loans with low documentation, as well as the overcollateralization percentage of the pool. Data on RMBS losses, downgrades, and price changes come from Bloomberg (as of 2014Q1). The results are presented for specifications with and without underwriter fixed effects for the top six underwriters in our sample. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Panel A: Pool Level Analysis: Second-Lien Misrepresentation, Losses, and Pricing						
	Losses		Yield spread		Subordination level	
	(1)	(2)	(3)	(4)	(5)	(6)
Percent misreported	0.284*** (0.057)	0.246*** (0.0599)	0.00415 (0.00446)	0.00344 (0.00471)	0.0306 (0.0222)	0.0205 (0.023)
Second Reported CLTV	0.687*** (0.1394)	0.724*** (0.1428)	0.0320*** (0.011)	0.0323*** (0.0113)	0.158** (0.0622)	0.165*** (0.0623)
Underwriter FE	No	Yes	No	Yes	No	Yes
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of pools	333	333	333	333	333	333
Mean	8.922	8.922	0.833	0.833	8.564	8.564
R ²	0.515	0.522	0.268	0.274	0.908	0.910

Panel B: Second-Lien Misrepresentation and Losses for Junior and Senior RMBS Tranches					
	Below-AAA losses (1)	Below-A losses (2)	AAA losses (3)	AAA ratings downgrades (4)	AAA price change (5)
Percent misreported	1.401*** (0.4109)	5.683*** (1.38)	0.214*** (0.0455)	-0.066*** (0.0199)	-0.448*** (0.1087)
Second Reported CLTV	2.446** (0.9784)	16.75*** (3.30)	0.787*** (0.1144)	-0.139*** (0.0368)	-1.893*** (0.2375)
Underwriter FE	Yes	Yes	Yes	Yes	Yes
Other controls	Yes	Yes	Yes	Yes	Yes
Mean	57.33	91.20	3.99	-17.24	-22.73
R ²	0.292	0.534	0.608	0.124	0.342

Table VI
Impact on RMBS Market: Which Investors Were Affected? Evidence from Holdings of Insurance Companies

This table examines a sample of RMBS tranches in our RMBS data set that were bought by insurance companies. Panel A shows the distribution of these tranches sorted by their initial rating from Standard & Poor's along with their cumulative average losses. Column (1) of Panel B presents OLS estimates in our main sample of 333 pools from a regression in which the dependent variable is a dummy that takes the value of one if a given pool has tranches that were bought by insurance companies, and zero otherwise. Column (2) of Panel B presents OLS estimates where the dependent variable is the pool's cumulative loss. The sample is restricted to only those pools whose tranches were bought by insurance companies (139 pools). Column (3) of Panel B presents OLS estimates where the dependent variable is the cumulative loss of tranches bought by insurance companies in the sample of 556 tranches bought by insurance companies. Column (4) of Panel B presents OLS estimates where the dependent variable is the change in the price of AAA tranches bought by insurance companies (in percentage terms) in a sample of 293 AAA-rated tranches bought by insurance companies for which Bloomberg reports these data. *Percent Misreported* corresponds to the percentage of loans in the pool having a second-lien misrepresentation (for a pool corresponding to the given tranches). *Reported CLTV* is the average reported CLTV of loans in the pool to investors. Other controls include the pool-level means of origination variables, such as borrowers' FICO scores or the fraction of loans with low documentation. Data on RMBS losses, ratings, and prices come from Bloomberg (as of 2014Q1), while data on insurance companies' holdings come from Merrill et al. (2014). The estimates are in percentage terms; standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Panel A: Distribution of RMBS Tranches Bought by Insurance Companies and Their Cumulative Losses			
Original rating	Number of tranches	Percent of total	Average loss
Aaa	403	72.48%	6.34%
Aa1	31	5.58%	66.23%
Aa2	20	3.60%	73.57%
Aa3	12	2.16%	93.56%
A1	6	1.08%	98.43%
A2	9	1.62%	93.94%
A3	6	1.08%	98.31%
Baa1	14	2.52%	99.60%
Baa2	9	1.62%	98.08%
Baa3	7	1.26%	100%
NR	39	7.01%	86.84%
Total	556	100%	28.07%

Panel B: Misrepresentations and RMBS Insurance Company Holdings				
	Whether bought by insurance company (1)	Losses on pools bought (2)	Losses on tranches bought (3)	Price change of bought AAA (4)
Percent misreported	0.0174*** (0.0050)	0.174** (0.069)	0.528** (0.2653)	-0.463** (0.2031)
Second	0.00945 (0.012)	0.612*** (0.1795)	2.917*** (0.6912)	-1.254*** (0.4511)
Origination CLTV				
Other controls	Yes	Yes	Yes	Yes
Mean	0.43	8.72	28.07	-27.80
R^2	0.245	0.249	0.315	0.337

5.7 Table VI: who held the affected securities?

Read:

- Insurance companies in the sample hold **556** RMBS tranches, of which **403** are AAA-rated.
- Average losses are substantial even for insurer-held AAA tranches (about **6.34%**) and enormous for lower-rated tranches.
- The paper finds no evidence that insurers avoided pools with larger shares of misrepresentation. If anything, those pools were somewhat more likely to be purchased by insurers.
- Within insurer-held securities, larger misrepresentation shares predict greater pool losses, greater tranche losses, and larger price declines on bought AAA tranches.

Economic message. Large institutional investors, not only the most junior specialist buyers, were exposed to and harmed by hidden collateral quality problems.

6 Short summary

- The paper identifies a direct, contractually important form of false disclosure in the RMBS market: undisclosed simultaneous second liens.
- Using matched BlackBox-Equifax data, it finds that about **7.13%** of loans reported as having no second lien are misrepresented, with true CLTV understated by about **19.6 percentage points**.
- These loans default far more often: about **10.15 percentage points** more within two years, which is roughly a **70%** higher default rate than comparable truthful no-second-lien loans.
- Internal lender files and deed records show that lenders typically knew about the second liens, so the misrepresentation likely happened late in the intermediation chain.
- Pools with more such misrepresentation suffered larger subsequent losses, and even AAA tranches in those pools faced larger losses, downgrades, and price declines.
- Yet the hidden risk was not reflected in standard issuance-pricing proxies, even though disclosed CLTV was strongly priced.
- The main estimates are conservative lower bounds, and additional evidence on owner-occupancy misreporting suggests broader RMBS collateral misrepresentation was even larger.

7 Conclusion

7.1 Core conclusion

The paper provides unusually direct evidence that major financial intermediaries sold RMBS backed by collateral whose quality was falsely described to investors. The main second-lien result is not an inference from bad outcomes alone; it is a direct comparison between contractual disclosure and independent credit-bureau records, backed by lender internal files and deed records.

7.2 Economic intuition

The economics are straightforward. Hidden junior liens reduce borrower equity, which sharply increases mortgage default risk. When that hidden leverage is omitted from investor-facing disclosure, pools look safer than they really are. Investors and rating agencies then price the security using disclosed CLTV rather than true CLTV, so intermediaries can sell riskier collateral on terms that would not prevail under truthful disclosure.

7.3 Takeaway

This paper shifts the crisis debate from “did investors underestimate macro risk?” to “were investors directly misinformed about the assets they bought?” The answer in this paper is yes. Moreover,

the effects are not small: they show up in defaults, pool losses, tranche losses, ratings, prices, and large-institution holdings. For studying securitization, disclosure, and intermediary incentives, this is one of the cleanest pieces of evidence that misrepresentation in a large debt market was both real and economically consequential.